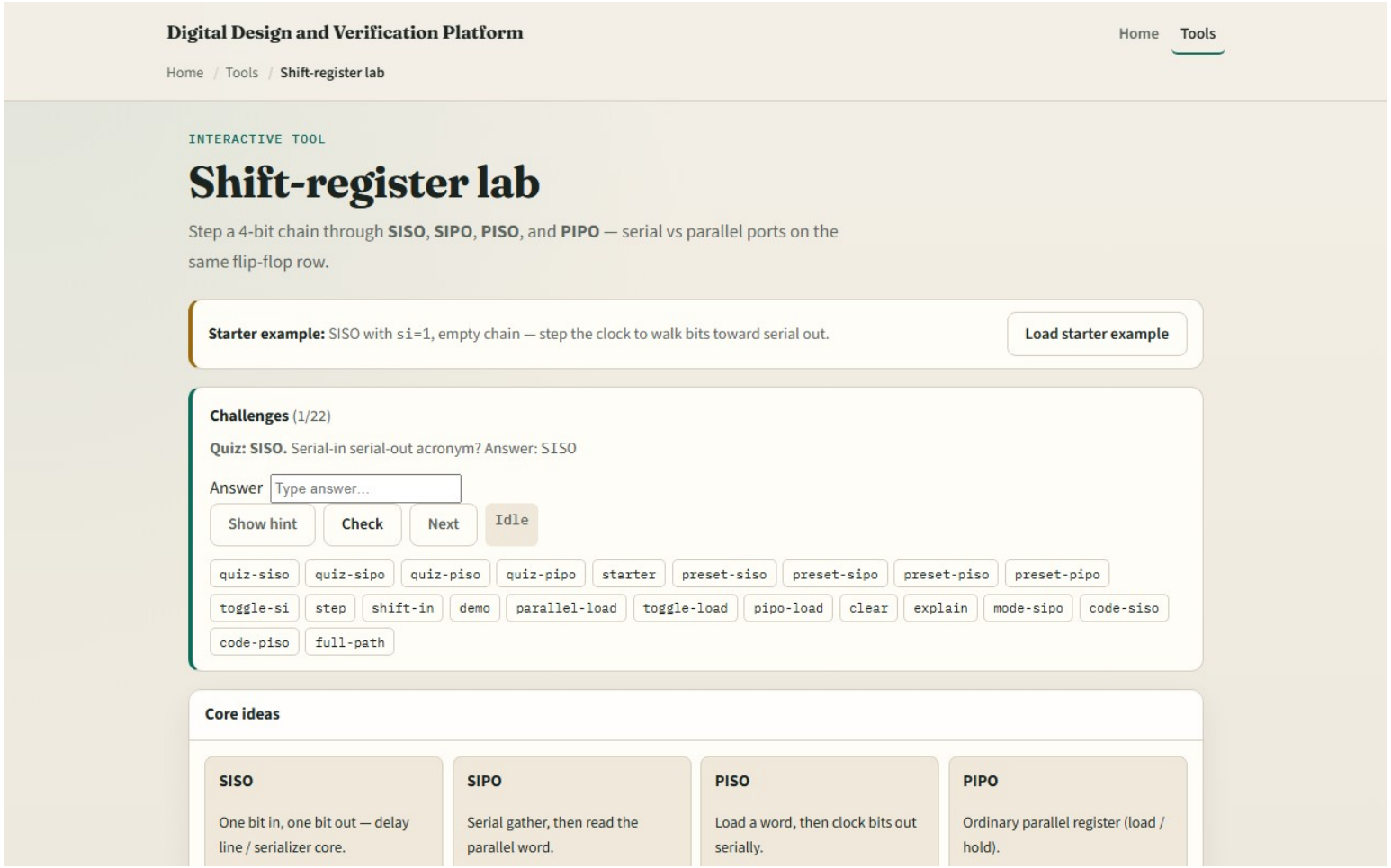
Shift-register patterns

A shift register is a chain of flip-flops where each stage passes its value to the next one on every clock edge

How shifting works

- On each rising edge, each flip-flop captures the output of the flip-flop to its left
- The leftmost stage captures the serial input
- After N clock cycles an N-bit shift register has moved each bit N positions to the right
- For parallel output, you read all stages at once
- For parallel load
- The ring version connects the last stage back to the first

Browser lab



Shift-register lab starter

Real Verilog practice

wsl – module15-shift-register-lab/examples/navigation

```
# Track A – real Unix
# real Verilog session (Track A)

$ cat examples/shift_demo/shift_demo.v
// shift_demo.v – SIPO and PISO shift registers
module sipo8(
  input        clk,
  input        rst,
  input        si,
  output reg [7:0] q
);
  always @(posedge clk) begin
    if (rst) q ≤ 8'd0;
    else     q ≤ {q[6:0], si};
  end
endmodule

module piso8(
  input        clk,
  input        rst,
  input        load,
  input  [7:0] d,
  output reg   so,
  output reg [7:0] shreg
);
  always @(posedge clk) begin
    if (rst)       shreg ≤ 8'd0;
    else if (load) shreg ≤ d;
    else           shreg ≤ {1'b0, shreg[7:1]};
  end
  always @(posedge clk)
    so ≤ shreg[0];
endmodule

$ iverilog -t null examples/shift_demo/shift_demo.v
(syntax check passed)
```

Real Verilog - shift register demo

Pitfalls to watch

- Do not use blocking assignment inside a clocked shift register
- Do not forget to account for the latency
- When connecting stages manually instead of using a concatenation
- And when using a ring, verify the feedback path or you may lose bits silently

Your turn

- Complete the checklist for at least one track, ideally both
- In the browser lab
- In real Verilog
- When you finish, take the short quiz